Imperial College London
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Causal Structure Discovery via a Self-Implemented FCI$^+$ Algorithm:

A Case Study on the Tennessee STAR Project

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IRP repository:
https://github.com/ese-ada-lovelace-2025/irp-yq225

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Abstract

This project presents an independent, auditable implementation of FCI and FCI+, validates the two algorithms, and applies them to the Tennessee STAR class-size experiment. FCI learns Partial Ancestral Graphs (PAGs) in the presence of latent confounding and selection, although its Possible-D-SEP search is computationally demanding; for sparse graphs, FCI+ instead admits a polynomial bound on conditional-independence (CI) queries. The resulting fully typed Python package has no causal-discovery dependency and provides both paper-aligned execution schedules and structured decision traces. Validation draws on 367 automated tests, three exact m-separation targets, a 15-run known-truth benchmark, and differential comparisons with causal-learn and R `pcalg`. Robust FCI+ achieves a mean skeleton F1 of 0.985 and endpoint accuracy of 0.840, compared with 0.977 and 0.737 for paper-profile FCI. Among 6,325 students in the STAR kindergarten cohort, small classes are associated with a 13.90-point observed kindergarten advantage (school-cluster 95% interval 5.21–22.07), while aide contrasts remain close to zero. Because the analysed outcomes are selected after assignment, this contrast is not an identified treatment effect. Across methods, skeleton relations are more defensible than endpoints that vary with ordering, threshold, or chronology; none of the learned PAGs identifies a unique DAG or an effect magnitude.

1 Introduction

Problem description. Many empirical studies pose causal questions even though some common causes remain unmeasured. Family background, institutional selection, and administrative recording can induce dependencies that cannot be faithfully represented by a causally sufficient DAG over the recorded variables; nevertheless fitting such a DAG converts genuine uncertainty into unsupported arrows. FCI is designed for precisely this setting. Under the causal Markov condition, faithfulness, and appropriate sampling assumptions, it uses conditional-independence (CI) information to learn a Partial Ancestral Graph (PAG) that accommodates latent confounding and selection [4, 8, 9]. This flexibility comes with an expensive Possible-D-SEP search and ambiguous endpoints. For sparse graphs with fixed maximum degree k, Claassen et al. [2] establish a polynomial worst-case bound on the number of CI queries.

This leads to two research questions. First, can FCI and FCI+ be reimplemented from their primary sources in a faithful and verifiable way? Second, what conclusions can that implementation legitimately support when applied to the Tennessee STAR class-size experiment, a real randomized educational study whose observed data are affected by attrition and selection?

Significance. Public FCI-family tools generally provide limited visibility into the evidence behind oriented endpoints. Among the interfaces examined in section 2, none offers a Python implementation of Claassen's FCI+ with paper-profile control and a structured audit trail. Linking each graph mark to recorded CI decisions supports defensible structural claims and differential testing. STAR also provides a canonical application in which selective attrition complicates longitudinal inference.

Objectives and contributions. Four measurable objectives address the auditability and finite-sample gaps identified above, with each objective also defining a contribution of the project.

O1/C1. *Implement source-faithful algorithms.* Success requires separate paper profiles whose stages map to primary-source concepts, explicit PAG endpoint semantics, typed public APIs, and no causal-discovery dependency. The delivered 9,559-line, 38-module package supplies five CI tests, ordered Possible-D-SEP, complete Algorithm 2 stages, and five orientation schedules.

O2/C2. *Verify and validate.* Success requires exact recovery of all pre-specified oracle targets, finite-sample accuracy reporting, and a clean automated test matrix. The apparatus comprises an exact MAG m-separation oracle, three published-structure targets, 15 known-truth runs, and 367 tests under strict typing on Python 3.9–3.13.

O3/C3. *Position against established software.* Success requires matched-input comparisons with disclosed interface mismatches, using skeleton, endpoint, runtime, and CI-query metrics. Executed causal-learn and R `pcalg` runners provide this evidence.

O4/C4. *Deliver a defensible STAR application.* Success requires independent R replication plus school-cluster bootstrap, cyclic-order, chronology, and sensitivity audits, with conclusions promoted only through a pre-specified evidence hierarchy.

C1 and C4 are separated so the algorithm remains domain-neutral while the application states its coding, selection, clustering, and interpretation choices. Section 2 defines the source and software gap; sections 3 and 4 evaluate O1–O4; and sections 5 and 6 distinguish supported conclusions from exploratory patterns.

2 Preliminaries and Related Work

2.1 Graphs, assumptions, and PAGs

Constraint-based discovery starts from statements of the form $X \perp Y \mid Z$ over observed variables V, while allowing latent variables L and selection variables S to be present. Under the *causal Markov condition*, graphical separation implies conditional independence; under *faithfulness*, observed independences arise from graphical separation rather than exact cancellation. In a DAG, the relevant separation criterion is d-separation:

Definition 1 (d-separation)**.** A path p is blocked by a node set Z if and only if p contains a chain or fork whose middle node is in Z, or a collider whose middle node is neither in Z nor an ancestor of Z. If Z blocks every path between X and Y, they are d-separated given Z [7].

When causal sufficiency holds, these assumptions justify DAG-based methods such as PC. Without causal sufficiency, the appropriate observed-variable representation is a maximal ancestral graph (MAG) [8]. Marginalising over L and conditioning on S can induce directed, bidirected, and undirected relations. Each missing edge in a MAG corresponds to an *m-separation*, which generalises d-separation to mixed ancestral graphs and provides the criterion used by the project's exact oracle. Because multiple MAGs may encode the same independence model, the identifiable output is a PAG that records precisely the endpoint features shared across a Markov equivalence class [12].

Every PAG adjacency has two endpoint marks. A circle leaves ancestry unresolved; an arrowhead at Y rules out Y as an ancestor of X; a tail asserts ancestry in every represented MAG; a bidirected edge is consistent with latent confounding but does not identify a particular latent variable; and an undirected edge reflects selection structure. An adjacency indicates only that the tested conditioning sets did not separate the pair. Neither an adjacency nor an endpoint by itself identifies a structural coefficient, an adjustment set, or an interventional contrast [7]. Because sample-based CI tests are susceptible to error, recovery under an exact oracle evaluates graph logic independently of finite-sample accuracy. Of the discovery families reviewed by Glymour et al. [4], score-based searches such as GES and standard functional causal models such as LiNGAM assume causal sufficiency. The constraint-based FCI family is therefore the appropriate choice when latent confounding and selection need to be represented.

2.2 Standard FCI and its historical boundary

FCI does not require every common cause of the measured variables to be observed [9, Ch. 6]. A PC-style search removes $X - Y$ when the pair is separated by some subset of their current neighbours. In the presence of latent variables, however, a separable pair may remain adjacent because a required separator can include a node that is no longer adjacent to either endpoint. The book uses D-SEP to characterise where an observed separator may lie and, because D-SEP cannot be computed from a partial graph, introduces an observable superset:

Definition 2 (Possible-D-SEP)**.** For an ordered pair (A, B) in a partially oriented graph, a node $V \neq A$ belongs to Possible-D-SEP(A, B) if some path from A to V has every internal triple either a collider or a triangle whose middle node is not a definite noncollider [9].

This construction is intentionally permissive and must be formed *separately* for each ordered endpoint. Searching in only one direction or combining the two pools changes the algorithmic contract. The complete schedule consists of a PC-style adjacency search, initial orientation of unshielded colliders, ordered Possible-D-SEP refinement, endpoint resetting and collider reconstruction from the final separators, and orientation

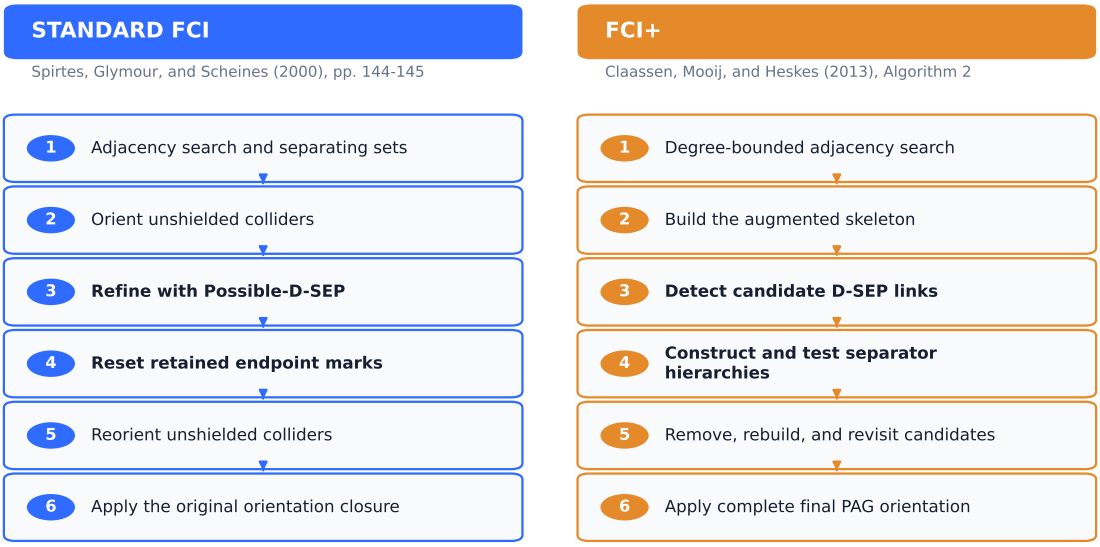


Figure 1: Source-located control-flow comparison. Standard FCI performs a non-local Possible-D-SEP refinement; FCI+ instead iterates augmented-skeleton, candidate D-sep-link, hierarchy, and minimal-separator stages before invoking final orientation. The robust application profile of section 3 changes finite-sample policy, not the definition of Algorithm 2.

closure. Given correct population-level CI decisions, Theorem 6.4 of the source guarantees a correct *partially oriented inducing path graph*. The authors nevertheless state that they did not know whether the result was maximally informative. Complete orientation therefore relies on the later rules of Zhang [12] and is not retrospectively attributed to the 2000 source.

2.3 FCI+: Algorithm 2

Claassen et al. [2] ask whether causal discovery remains intrinsically exponential when the observed structure is sparse. Their central result, restated in the form used throughout this report:

Theorem 1 (2)**.** Assume a constant-time exact independence oracle, a distribution faithful to a causal DAG, and an observed-variable MAG of maximum degree at most a constant k. Then Algorithm 2 (FCI+) returns the completed PAG over N observed variables using $O(N^{2(k+2)})$ CI queries in the worst case.

The scope of this result is narrower than the paper's title suggests: it bounds oracle queries, but does not render a statistical test exact or constant-time. A pair that is nonadjacent in the true MAG may survive a bounded adjacency search if every useful separator contains a node outside the union of the endpoints' adjacencies. Such a spurious adjacency is called a *D-sep link*. Instead of searching a large Possible-D-SEP pool, FCI+ encodes invariant non-ancestry in an *augmented skeleton* G^+: when a known minimal independence $X \perp Y \mid S$ becomes dependence given $S \cup \{A\}$ for a suitable adjacent A, an invariant arrowhead at A is supported. In G^+, a D-sep link $X - Y$ appears as the middle edge of $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$ with U, V nonadjacent and cross-direction paths not running against an arrowhead; the pattern only *proposes* a candidate, and independence evidence determines whether deletion is warranted. Conditioning sets are drawn from the recursive hierarchy $\mathrm{HIE}(X, I)$, defined as the fixed point obtained by repeatedly adding the recorded minimal separators of its members. Since candidates may depend on one another, each successful removal rebuilds the augmented skeleton and candidate list (figure 1). The arXiv supplement's abstract states $N^{2(k+1)}$, but its theorem and proof agree with the UAI paper's $N^{2(k+2)}$. Algorithm 2 invokes complete FCI orientation, so its completed-output claim depends on Zhang [12].

Table 1: Dated public-interface capability comparison (26 July 2026). All four tools expose standard FCI; "partial" means some information is obtainable, but not through the same integrated result contract. Tetrad is documentation-only because no working Java runtime was available.

Tool	Claassen FCI⁺	Custom CI	Sepset provenance	Order/bootstrap audits	Structured audit and export
`fci_engine` 0.1.0	yes	yes	yes, stage named	both	CI and orientation traces; D-SEP diagnostics; per-edge explanations; JSON/CSV/HTML
`pcalg` 2.7-12	yes	yes	partial	order partial	standard-FCI sepsets; no equivalent bundled export established
causal-learn 0.1.4.7	no public entry point found	named tests	not exported	not established	graph return; no equivalent structured trace established
Tetrad 7.6.10	not established	yes	yes	partial, broader workflow	logger and GUI facilities; not executed here

2.4 What the sources do not establish

Neither source guarantees recovery in finite samples, since both sets of theoretical results assume correct CI decisions. Neither source identifies a unique DAG, an adjustment set, or a treatment effect, and a bidirected pattern neither names nor measures a hidden cause. The 2000 source establishes neither maximal orientation nor a polynomial bound. The 2013 source, in turn, establishes no finite-sample superiority and offers no guarantee when k is misspecified: a user-supplied bound is not evidence that the assumption holds. Its authors also note that hierarchy-derived conditioning sets can be large, thereby reducing power in finite samples.

2.5 Existing software

Table 1 summarises the dated capability comparison: `pcalg` [5] exposes both algorithms and accepts user-defined CI tests, but its `fciPlus` result carries materially less separator provenance; causal-learn [3] exposes standard FCI but no public Claassen FCI⁺ entry point; Tetrad [10] provides FCI-family hybrids, none established as a drop-in Claassen FCI⁺, and was checked at documentation level only (no working Java runtime). The originality of this work is therefore not a new algorithm but a verifiable one: paper-profile control, structured traces, separator provenance, per-edge explanations, and stability auditing integrated in one workflow — an auditability improvement, not a universal accuracy or speed claim.

3 Methodology

3.1 Package design

The package translates the requirements set out in section 2 into inspectable components while keeping algorithmic implementation separate from application policy. Reusable code resides in `src/fci_engine`, the STAR workflow in `case_studies/tennessee_star`, and the estimator contains no STAR-specific choices. The two algorithms have separate entry points and explicit paper profiles, preventing a finite-sample default from silently substituting for the source algorithm. `FCIConfig.paper()` specifies unbounded depth, immediate deletion, first-found separators, enabled Possible-D-SEP, nonconservative colliders, and the historical `spirtes_2000` closure; `FCIPlusConfig.paper(k)` requires an explicit k for both conditioning depth and sparsity bound, disables Possible-D-SEP, and selects the complete `standard` closure required by Algorithm 2's output claim. These profiles govern execution but do not validate theorem assumptions; setting $k = 3$ does not establish the true MAG's degree.

The `PAG` class stores a mark at both ends of every adjacency, explicitly representing tails, arrowheads, circles, and absence. All searches share the `CITest.test(data, x, y, cond_set)` interface, with built-in Fisher-Z, missing-value Fisher-Z, chi-square, G-square, and kernel tests. Cache canonicalisation distinguishes logical CI requests from cache hits and executed statistics, as required when counting oracle queries.

`FCI.fit` makes the source-prescribed stage order visible and labels Possible-D-SEP removals with the

source `pdsep`. `FCIPlus.fit` intentionally avoids calling Possible-D-SEP, passing the initial skeleton instead to the Algorithm 2 sequence: augmented-skeleton construction, bidirected witness patterns, paired endpoint bases, hierarchy fixed points, separator testing and minimisation, graph updating, and candidate rebuilding. Its diagnostics count candidates, revisits, hierarchy and D-SEP queries, and the largest conditioning set; these are accounting measures, not a proof of the bound. A single dispatcher provides five orientation schedules: `spirtes_2000`, the historical closure used by the standard-FCI paper profile; `standard`, the modern complete schedule required by the FCI⁺ paper profile; and `conservative`, `leaf`, and `robust`, finite-sample engineering policies supporting no completed-PAG claim. A regression verifies that `spirtes_2000` stops before a selection-bias tail completion that `standard` orients, making the historical boundary executable. `FCIResult` binds the final graph to the evidence that produced it — configuration, separators with stage sources, logical CI counts, and named CI and orientation traces; `explain_edge` returns every decision behind a pair, so a plotted arrow cannot detach from the profile, test, threshold, and rule that created it.

The project comprises a 9,559-line, 38-module algorithm package, 367 tests, a 3,052-line STAR application, an independent 396-line R runner, and supporting scripts. It includes `py.typed` and passes Ruff, strict MyPy, wheel building, and Pytest on Python 3.9–3.13. These checks establish interface and packaging consistency, not source faithfulness, sparsity, or empirical conclusions.

The practical profile exchanges additional computation for greater stability. Stable deletion reduces order dependence; max-p separators use more CI queries to seek stronger independence evidence; conservative collider and tail policies retain circles; and automatic α follows a documented sample-size heuristic. Consequently, "paper FCI⁺" is evaluated for source fidelity and query efficiency, whereas "robust FCI⁺" is the preferred policy for the application because it satisfies the stated audit criteria while producing less-specific endpoints. This is a preference for the present application, not a universal ranking.

3.2 Validation and comparison design

The most stringent tests replace sample-based CI decisions with exact m-separation in a known MAG, thereby separating search correctness from sampling error. Three independently specified targets are based on published structures: the canonical Figure 4(b) D-SEP example of Claassen et al. [2], a latent-variable reference graph from the FCI literature, and an orientation-rule reference graph associated with Zhang [12]. Finite-sample behaviour is subsequently evaluated on 15 deterministic latent-SEM cases: five graph families, three seeds per family, 2,500 observations, and known oracle PAGs.

The executed software comparison evaluates identical committed data under recorded settings. The benchmark reuses the 15 known-truth runs, every method receiving the same numeric matrix and variable order. Fixed paper and reference rows use Fisher-Z at $\alpha = 0.001$; robust FCI⁺ uses its documented sample-size rule, which resolves to $\alpha = 0.01$ here — intentional: that row evaluates the public robust policy, not a literal Algorithm 2 reproduction. Exact F1 requires both endpoint marks to match; semantic F1 additionally credits endpoint patterns compatible with the target ancestral constraints; endpoint accuracy scores individual marks. The common conversion represents each edge as an unordered pair with two semantic marks. Where interfaces permit, data, order, alpha, and CI semantics are fixed; remaining interface mismatches are disclosed.

3.3 STAR data and panels

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment spanning kindergarten to grade 3. It assigned students and teachers to small classes (approximately 13–17 students), regular classes (approximately 22–25), or regular classes supported by a full-time aide [11]; it remains a central source of evidence on class size [6]. This analysis uses the official Harvard Dataverse release [1]: 11,601 student records, 379 variables, CC0 licence, codes checked against the 146-page user guide.

Because randomization provides a stronger basis for estimating the class-size effect than structure learning, FCI is not used to rediscover that effect. Instead, the analysis addresses structural questions: the attrition pattern in grade-3 observation, persistence of the early–later achievement adjacency, the consequences of including an early post-assignment outcome, agreement across implementations, and the search savings of FCI⁺. The experimental design also provides an external audit. If a data-only PAG implies an implausible

Table 2: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate assignment, characteristics, context, and kindergarten achievement to whether both grade-3 scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

direction into a pre-treatment variable or latent confounding of randomized assignment, the result is treated as a discrepancy rather than interpreted literally.

The analysis includes 6,325 students with a kindergarten class assignment across 79 schools. They are organised into three panels that are deliberately kept separate because each represents a different statistical question (table 2). Every model input is an integer-coded discrete category: class arm, sex, ethnicity (with small source categories combined), free-lunch status, school context, entry-age quantiles, binned teacher experience, and reading-plus-mathematics scaled scores transformed into achievement quartiles. The exact numerical tables supplied to the algorithms are committed to the repository.

3.4 Discovery protocol and audits

Before structure discovery, randomized-arm contrasts are calculated for kindergarten scores, grade-3 scores among students for whom both scores are available, and the probability that both scores are observed. Uncertainty is estimated from 1,000 kindergarten-school bootstrap samples, and these design-based contrasts are reported separately from the discovery results.

Discovery uses the likelihood-ratio G-square test, $G^2 = 2\sum_s \sum_{x,y} O_{xys} \log(O_{xys}/E_{xys})$, at $\alpha = 0.05$ — chosen because every panel variable is an integer-coded category, where Fisher-Z's Gaussian-linear assumptions do not apply, at the cost of sparse contingency cells under high-order conditioning. Standard FCI runs its unbounded paper schedule; paper FCI$^+$ uses $k = 3$; robust FCI$^+$ uses the practical profile (stable skeleton, strongest separating sets, conservative colliders, robust orientation, depth and sparsity bound 3); the independent reference is `pcalg::fciPlus` with selection bias enabled and no exposed k. No temporal priors are supplied to discovery; reversals are audited after fitting.

Alternative choices were excluded for specific reasons. PC and GES require causal sufficiency; Fisher-Z is poorly matched to the discrete panels; kernel CI is substantially more expensive and does not address clustered sampling; and temporal priors would mask chronology failures rather than reveal them. A formal model of the randomized treatment effect complements the structural question and is therefore kept separate from the discovery analysis.

Stability is examined through three audits. First, 100 school-cluster bootstrap refits are performed for each algorithm and panel, recording how often each unordered adjacency appears; this frequency is an exploratory diagnostic, not a posterior probability. Second, a cyclic-order audit refits all profiles and the R reference under every cyclic shift of the variable list. Third, the focused panel is evaluated over a sensitivity grid with $\alpha \in \{0.01, 0.05\}$, three or four quantile bins, and $k \in \{2, 3, 4\}$.

A conclusion is promoted at the skeleton level only if the adjacency survives every cyclic shift and has a high bootstrap frequency. A directional claim must additionally have invariant endpoint marks, agree with measurement time and randomization, and show no material reversal in sensitivity analyses. If these conditions fail, direction is reported as unresolved rather than the edge being discarded.

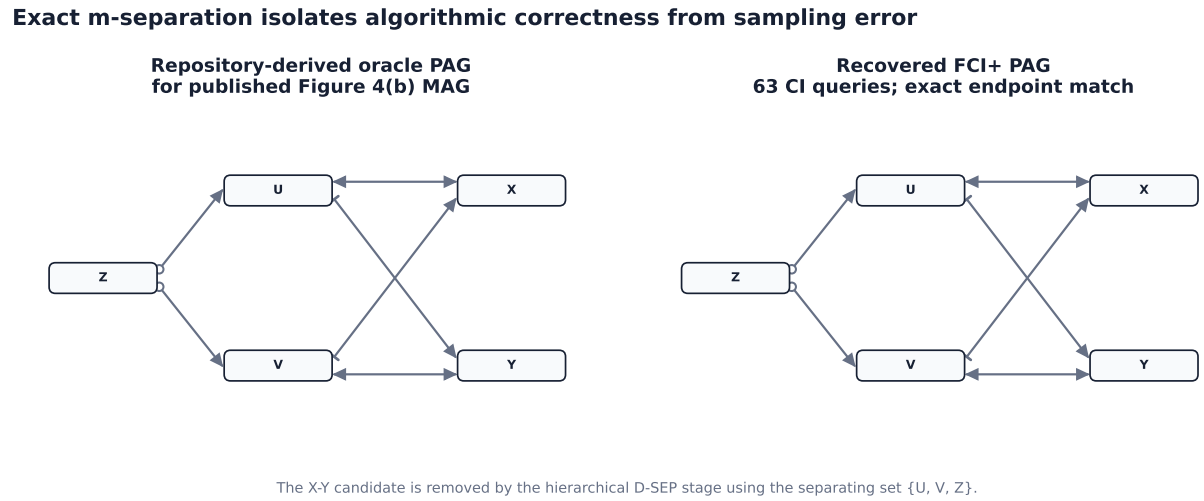


Figure 2: Exact-oracle validation on the central Figure 4(b) example. The source-grounded facts are the six MAG edges and the exhibited separator $\{U, V, Z\}$; the endpoint-complete comparison target is independently derived in the repository. Paper-aligned FCI needs 102 oracle queries, FCI+ 63; both recover the target exactly.

Table 3: Executed known-truth comparison: means over 15 completed runs (five latent-SEM families, three seeds each, 2,500 observations).

Method (version)	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	Seconds	Logical CI
Local FCI paper (0.1.0)	0.977	0.532	0.943	0.737	0.020	449.7
Local FCI+ paper (0.1.0)	0.979	0.536	0.951	0.744	0.009	197.1
Local FCI+ robust (0.1.0)	0.985	0.730	0.985	0.840	0.018	401.9
causal-learn FCI (0.1.4.7)	0.979	0.536	0.951	0.744	0.009	not exposed
R `pcalg` FCI+ 2.7-12	0.979	0.536	0.951	0.744	0.500	not exposed

4 Results

4.1 Exact-oracle validation

Both algorithms recover each of the three encoded targets. Figure 4(b) is also evaluated under four variable orders. Its MAG contains six edges ($Z \to U$, $Z \to V$, $U \leftrightarrow X$, $V \to X$, $U \to Y$, $V \leftrightarrow Y$); the adjacency stage retains a false $X - Y$ link although $X \perp Y \mid \{U, V, Z\}$, with Z adjacent to neither endpoint. The source presents this separator but does not provide an endpoint-complete PAG; the completed comparison target is therefore derived in the repository (figure 2). In this example, paper-aligned standard FCI requires 102 CI queries and FCI+ requires 63, a reduction of 38.2%, while both recover the exact target. The values, target hash, and separator provenance are frozen in `reports/data/oracle_validation_summary.csv`.

4.2 Finite-sample known-truth benchmark

On a seeded latent linear SEM over the same reference graph, at $N = 5{,}000$ FCI+ returns an extra $X \leftrightarrow Y$ (exact-edge F1 0.923) while standard FCI recovers the exact PAG; at $N = 50{,}000$ both are exact — a reproducible fixture, not a universal threshold.

Across the 15 cases (the first three rows of table 3), the robust profile achieves higher mean skeleton and endpoint scores at approximately 2.04 times the CI cost of paper FCI+. This is not universal superiority, but shows that the STAR stability gain is not obtained simply by discarding structure. Fewer CI tests need not improve accuracy because the algorithms follow different paths after the initial skeleton. Finite-sample results remain sensitive to CI error and tuning despite exact-oracle validation.

Table 4: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples. Grade-3 summaries condition on both scores being observed; kindergarten summaries also exclude missing score components, so both are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Outcome	Arm	*n*	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

4.3 Comparison with established software

The identical aggregate scores for local paper-FCI⁺, causal-learn, and `pcalg` provide differential evidence, not equivalence. Local timings exclude process startup, whereas R includes it, and `pcalg::fciPlus` lacks an identical degree-bound control. The comparison therefore demonstrates matched behaviour only where interfaces overlap.

4.4 Observed randomized-arm outcomes

At kindergarten, the small-minus-regular contrast is 13.90 points (school-cluster 95% interval 5.21 to 22.07), compared with 0.31 points (−7.40 to 7.38) for regular-with-aide versus regular. Among students observed at grade 3, the small-class contrast remains 11.69 points (3.74 to 19.38), while the aide contrast is 0.51 (−5.70 to 7.22). The observation rates do not show a precise difference by arm: small versus regular is +1.54 percentage points (−1.73 to 4.56), and aide versus regular is −2.05 (−5.35 to 1.41) (table 4).

4.5 Structure discovery across the three panels

For the attrition panel, the three implementations show substantial agreement at the skeleton level (figure 3; costs and agreement are reported in table 5). None retains a direct `K_Class`–`Grade3_Observed` adjacency, which appears in only 12–16% of school bootstraps. By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of bootstrap fits for all three local profiles, and the free-lunch/observation adjacency is likewise stable. Thus, later data availability is not independent of early achievement and measured context. The endpoint interpretations nevertheless diverge. Standard FCI produces the impossible *grade-3 observation → kindergarten achievement*; self-FCI⁺ retains a circle; and R-FCI⁺ produces the temporally admissible direction. Although the R result is more plausible, software identity cannot establish correctness. Nor does the PAG prove a missingness mechanism, which requires an explicit estimand and additional assumptions.

In the longitudinal panel, every implementation retains the kindergarten/grade-3 achievement adjacency, which also appears in all 100 bootstrap samples for each local profile. None of the full-data PAGs retains a direct class-assignment/grade-3 adjacency once kindergarten achievement is included, yet that adjacency reappears in 71% of FCI, 79% of paper FCI⁺, and 80% of robust-profile bootstrap fits. Thus order invariance does not eliminate sensitivity to school resampling. Both FCI⁺ implementations orient grade-3 achievement toward kindergarten achievement, which is temporally impossible if read causally, while standard FCI leaves that endpoint unresolved. Agreement cannot override chronology because both may reproduce the same CI pattern under complete-case selection.

Excluding kindergarten achievement changes the result. All three algorithms return the same 10-edge PAG, rather than merely the same skeleton, including *K_Class* ↔ *Grade3_Achievement* (figure 4), an adjacency

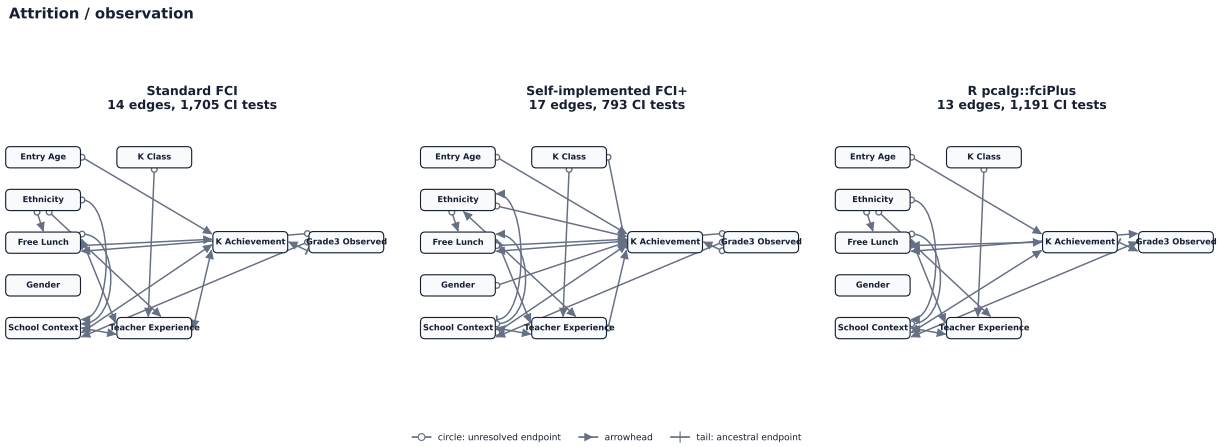


Figure 3: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI+, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs across implementations.

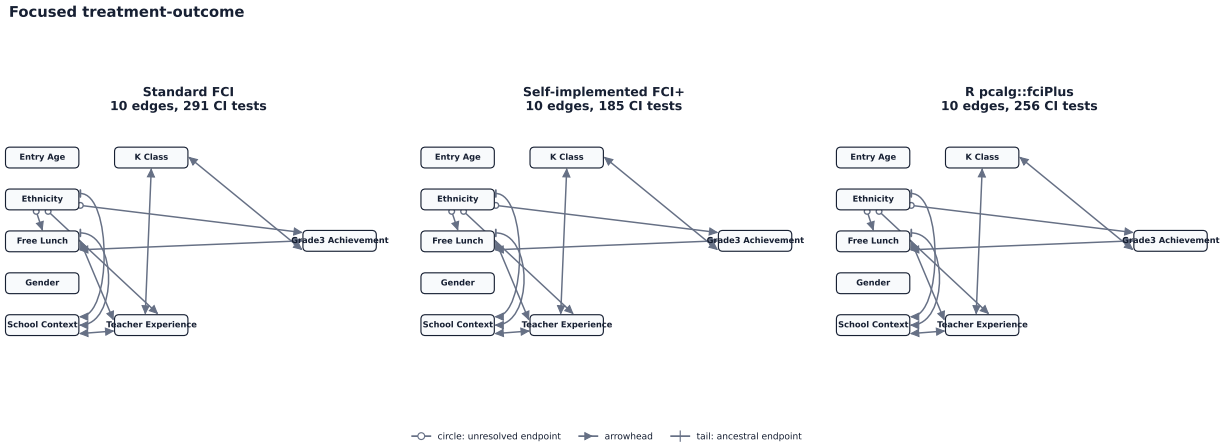


Figure 4: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

appearing in 92%, 85%, and 96% of the FCI, paper, and robust bootstraps, respectively. Exact replication strengthens this conditional-independence pattern but does not establish latent confounding. Kindergarten class assignment was randomized, whereas the focused panel conditions on observed grade-3 outcomes and omits early post-assignment achievement. Selection on observation, school-level dependence, omitted post-randomization variables, and CI-test error could all generate arrowheads at both endpoints.

The sensitivity analysis gives a clear result (figure 5). The focused adjacency is stable across the tested bin counts and degree bounds, but not across CI thresholds. At $\alpha = 0.01$, no local configuration retains the adjacency. At $\alpha = 0.05$, the paper profiles produce a bidirected mark for every tested k, whereas the robust profile retains circles at the endpoints. Because the stricter threshold requires stronger evidence against independence, deletion at $\alpha = 0.01$ suggests that the edge reflects comparatively weak conditional-dependence evidence rather than an invariant feature.

4.6 Robust application profile, agreement, and cost

The robust profile is a finite-sample decision layer, not a fourth paper implementation. Across the three panels it returns 17, 12, and 10 edges, with target relations *K_Achievement* $\circ\!-\!\circ$ *Grade3_Observed*, *K_Achievement* $\circ\!-\!\circ$ *Grade3_Achievement*, and *K_Class* $\circ\!-\!\circ$ *Grade3_Achievement*. It preserves target skeletons while avoiding the chronology reversal and over-interpretation of the bidirected edge.

In the cyclic-order audit, paper self-FCI+ reproduces the exact full-data PAG for 9 of 9 attrition shifts, 5 of 9 longitudinal shifts, and 6 of 8 focused shifts, including one temporal reversal in the longitudinal

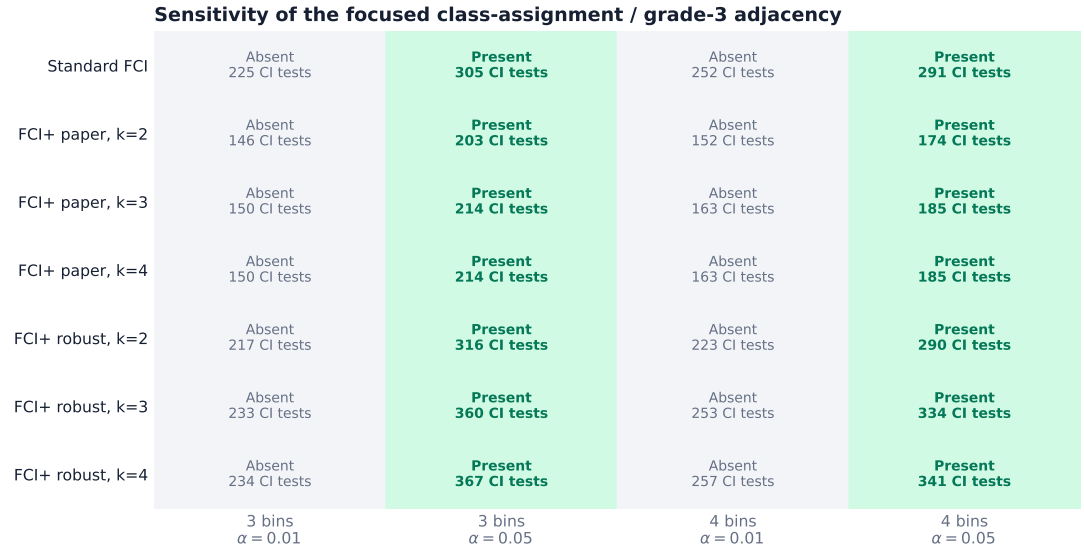


Figure 5: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins or $k = 2, 3, 4$ is used for FCI⁺.

Table 5: Three-algorithm structural agreement (left) and search cost (right). Endpoint rates use only adjacencies shared by the indicated pair; time is the median of three full-data repetitions.

Panel	Pair	Jaccard	Endpoints
Attr.	FCI / self	0.824	50.0%
	FCI / R	0.929	61.5%
	self / R	0.765	38.5%
Long.	FCI / self	0.769	50.0%
	FCI / R	0.833	70.0%
	self / R	0.923	83.3%
Focus.	FCI / self	1.000	100.0%
	FCI / R	1.000	100.0%
	self / R	1.000	100.0%

Panel	Algorithm	Edges	CI	Sec.
Attr.	FCI	14	1,705	5.437
	self paper	17	793	1.788
	self robust	17	1,304	2.838
	R-FCI⁺	13	1,191	10.394
Long.	FCI	10	960	2.148
	self paper	13	335	0.337
	self robust	12	619	0.562
	R-FCI⁺	12	475	1.650
Focus.	FCI	10	291	0.322
	self paper	10	185	0.159
	self robust	10	334	0.247
	R-FCI⁺	10	256	0.884

panel. The R reference preserves the skeleton for every panel and exactly matches endpoints in 7 of 9, 6 of 9, and 6 of 8 shifts, respectively. Robust self-FCI⁺ returns an identical PAG in all 26 refits without a reversal. Table 5 summarises both structural agreement and search cost. Paper FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard-FCI CI-test counts on the three panels (R reference: 69.9%, 49.5%, 88.0%). Runtime is not language neutral, so CI-call counts are the more interpretable comparison. Relative to paper self-FCI⁺, the robust profile uses 1.64–1.85 times as many CI calls, exchanging speed for stable deletion and invariant conservative endpoints. Overall, it returns the same complete PAG under every tested order, retains every target adjacency, and introduces no backward-in-time directed edge. These gains come at the cost of additional CI work and less orientation. Reliability is improved, but identifiability is unchanged.

5 Discussion

5.1 Evidence hierarchy

The evidence hierarchy prevents a definite PAG mark from outweighing the experimental design. Tier 1 is randomized assignment with a school-cluster contrast. Tier 2 comprises adjacencies replicated across methods, resamples, orders, and sensitivity settings, supporting skeleton-level claims but not magnitude or direction.

Tier 3 contains stable skeletons with circles or endpoint disagreement. Tier 4 includes implementation-, order-, or chronology-dependent endpoints and is diagnostic only. Tier 5 contains threshold-sensitive or complete-case patterns requiring external validation.

5.2 What can and cannot be concluded

The Tier-1 result, a 13.90-point small-class contrast, is consistent with an early benefit. It is not, however, an identified causal effect within the post-assignment observed subset without further assumptions about missing outcomes, compliance, and the original randomization blocks. Aide contrasts are close to zero at both time points, providing no evidence that an aide reproduces the small-class advantage. The achievement–achievement and achievement–observation adjacencies recur across all fits and bootstraps, supporting a robust *skeleton-level* relationship among early performance, later performance, and later availability (Tier 2). The full-data attrition PAGs contain no direct class-arm/observation adjacency, and its bootstrap frequency is low. Concern about selective attrition is therefore supported without evidence of a stable direct arm effect on observation.

Several conclusions remain unsupported. No unique DAG is identified because each PAG represents an equivalence class, and FCI supplies no effect magnitude. A bidirected edge does not prove the existence of a hidden confounder; in the focused panel, such an interpretation conflicts with known randomization unless selection or post-treatment conditioning is taken into account. Temporal reversals are diagnostics of the algorithm rather than credible chronology. Agreement across implementations is also not proof, since shared CI decisions and rules can reproduce the same incorrect endpoint. The 26 recorded shifts constitute a finite audit rather than a proof of order invariance, and bootstrap frequency is not a posterior probability. Finally, absence is conditional on policy: failure to retain an edge under one node set, test, and threshold does not demonstrate a zero effect.

5.3 Limitations of the method

Guarantees for the FCI family depend on faithfulness and correct CI decisions, as illustrated in practice by the threshold-sensitive edge in the focused panel. Because the true degree of the STAR MAG is unobserved, $k = 3$ is an analysis choice rather than a verified sparsity condition. Discretisation makes G-square testing possible, but loses within-bin information and can produce sparse conditional tables. The row-wise CI test is not robust to clustering. School-level bootstrapping reveals sensitivity to school composition, but does not correct this mismatch. Finally, version 0.1.0 and the manually specified oracle targets provide strong regression evidence, not a formal proof of correctness or an exhaustive check of MAG equivalence.

A deterministic scaling audit, covering 5 to 80 nodes in both a directed chain and the Figure 4(b) core with additional isolated variables, recovers the exact target skeleton in every run. It also reveals that the query advantage is not universal. FCI$^+$ performs better on the five-node core (62 versus 101 queries), but uses 5,162 queries versus 3,251 at 80 nodes and 24,338 versus 12,638 on the 80-node chain. This is consistent with a worst-case bound: augmentation carries a genuine cost when targeted D-SEP search offers little benefit.

5.4 Limitations of the STAR study

Both the longitudinal and focused panels condition on observed grade-3 scores, which may open selection paths. Kindergarten achievement is measured after assignment: including it may block part of the class-size pathway, whereas excluding it may leave post-treatment dependence. Presenting both panels makes this modelling choice visible but does not establish either model as correct. The analysis does not reconstruct the original randomization blocks, class switching, compliance, or school-specific procedures. Moreover, the composite score leaves subject-specific and subgroup effects unexplored.

6 Conclusion and Future Work

All four objectives were achieved, subject to clearly stated limits. **O1 was met**: the package provides source-located profiles of FCI and Algorithm 2 with typed, auditable outputs. **O2 was met**: it recovers all three exact targets, passes 367 tests, and quantifies finite-sample error through the 15-run benchmark. **O3 was met within the available interfaces**: matched causal-learn and `pcalg` runs reproduce the aggregate paper-profile scores, although neither exposes every control or diagnostic. **O4 was met within its structural**

scope: the STAR workflow completes all planned replication and stability audits, while identifying neither a formal intention-to-treat effect nor a unique causal graph.

The STAR evidence supports a 13.90-point observed kindergarten advantage for small classes, no comparable benefit from an aide, a stable early–later achievement structure, and selective later observation. Nevertheless, the independently replicated 10-edge focused PAG remains threshold-sensitive, and some endpoints vary with ordering or conflict with chronology. By retaining circles, robust FCI$^+$ avoids these literal reversals at 1.64–1.85 times the CI cost of the paper profile. This improves reliability without improving identifiability.

The main engineering challenge was translating ordered Possible-D-SEP pools and Algorithm 2's hierarchy and revisit semantics into executable invariants. Oracle-first validation, separation of algorithm from application, and committed seeds and artifacts proved most effective. Literal paper profiles produced unstable endpoints on real data, so robust FCI$^+$ is preferred for this application; paper FCI$^+$ remains the fidelity and query-efficiency reference.

The central contribution is an auditable account of conditional-independence constraints and modelling sensitivity, rather than a rediscovered causal model of STAR. Future work should reconstruct the randomization strata for a formal intention-to-treat analysis; introduce cluster-aware CI tests and endpoint-level stability summaries; analyse reading and mathematics separately; evaluate temporal background knowledge in a distinct analysis; and derive target PAGs using an exhaustive MAG-equivalence oracle.

Reproduction. The `run_case_study` module rebuilds the STAR application with three benchmark repetitions, 100 discovery bootstraps, and 1,000 descriptive bootstraps; its refresh option re-executes the R reference. Scripts under `reports/` regenerate every figure, the frozen Figure 4(b) evidence, and the scaling audit. The committed `STAR_Students.tab.gz` has SHA-256 `2ffa578822eb30fcd6626fa7c5cc734a721fd05dd522403f0d643f89e891bac8`.[1]

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[1] This document is the standalone final report; every number is regenerated from committed artifacts by the listed scripts. Extended per-stage evidence tables and complete source locators are kept in the repository's full report, reports/fci_plus_star_report.tex.